Yu Voon Ng

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Research Interests

Galaxy formation and evolution (including CGM and ISM). Large-scale structure.

Education

Max Planck Institute for Extraterrestrial Physics , Ph.D. in Astrophysics — Sep 2026 – PRESENT

National Taiwan University, M.S. in Astrophysics — Sep 2023 – Jun 2025

- Thesis: Probing the Baryon Cycle in the Local Universe with DESI Year 1 Data
- Advisor: Prof. Ting-Wen Lan

National Taiwan University, B.S. in Mathematics — Sep 2017 – Jun 2021

Research Experiences

Clustering Redshift — Sep 2025 – Jul 2026

- Advisor: Dr. Yi-Kuan Chiang
- Research topic: Optimized, packaged, and documented the clustering-redshift code Tomographer for public release
- Achieved a 4$\times$ improvement in computational efficiency and 10$\times$ reduction in memory requirements, enabling fast clustering-redshift estimation on standard personal computers
- Validated the clustering-redshift estimator against systematic effects, assessing its robustness for large-scale survey applications

The Circumgalactic Medium of DESI Bright Galaxies — Sep 2023 – Jul 2025

- Advisor: Prof. Ting-Wen Lan
- Research topic: Explored the relationship between galaxy properties and the cool CGM using DESI Year 1 data
- Characterized CGM absorption as a function of stellar mass, redshift and galaxy types, identifying trends linked to galactic feedback and redshift evolution
- Optimized data extraction and normalization of quasar spectra through parallel computation
- Conducted stacking analysis on normalized quasar spectra, enhancing sensitivity to m$\mathrm{\AA}$ level

The Metallicity of DESI Dwarf Galaxies — Aug 2022 – Jul 2023

- Advisor: Prof. Ting-Wen Lan
- Research Topic: Studied the metallicity of low-mass galaxies from DESI datasets using the direct method
- Increased the sample size of low-mass galaxies by 3 times, enhancing constraints on the mass-metallicity relation at the low-mass end
- Implemented a Convolutional Neural Network (CNN), achieving 99.15% purity in identifying dwarf galaxies from DESI imaging data

Publications

Tomographer: End-to-end Redshift Distribution Estimation for Source Catalogs and Intensity Maps
Chiang, Y.-K., **Ng, Yu Voon**, Lin, Y.-R., Taghizadeh-Popp, M., Ménard, B., 2026, submitted

A Comprehensive Characterization of Galaxy–cool CGM Connections at $z < 0.4$ with DESI Year 1 Data
Ng, Yu Voon, Lan, T.-W., Prochaska, J. X., Saintonge, A., Chang, Y.-L., Siudek, M., et al., 2025, ApJ, 993, 92

Collaborations

Dark Energy Spectroscopic Instrument (DESI) | Member — Aug 2022 – Jun 2025

Oral Presentations

Celebrating 40 Years of Damped Lya Systems	18 Mar 2026
10th Galaxy Evolution Workshop	9 Aug 2024
Spotlight Talks in DESI Summer Meeting	10 Jul 2024
ASROC Annual Meeting 2024	1 Jun 2024
Taipei Astronomy Workshop	8 Jan 2024
ASROC Annual Meeting 2023	19 May 2023

Awards and Honors

Dean's Award *(top 10% of class)*	2025
Donated Scholarships for Overseas Chinese Students	2025
Certificate of Merit for Outstanding Academic Performance and Good Behavior	2021
Scholarship for Overseas Chinese Students with Outstanding Academic Performance	2020

Teaching Assistant

General Astronomy (3 credits, $\sim$100 students) — Spring 2024
Supported instruction through grading, student mentoring, and contributing to lectures and exams

Seminar ($\sim$90 students) — Spring 2025
Supported seminar coordination through poster design, student communication, and facilitating presentations

Skills

Programming: Python, SQL, C/C++

Data Analysis: Parallel Computing, Machine Learning

Software: RADMC-3D, FARGO3D, ROOT, CASA, Miriad, SAOImage DS9

Coursera: Google Data Analytics, Introduction to Statistics

Language: Chinese, English, Malay